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Instruction

What the question is

Regular Expression Practice: 1.18

1.18 Give regular expressions generating the languages of Exercise 1.6.

1.6 Give state diagrams of DFAs recognizing the following languages. In all parts, the alphabet is $\{0,1\}$.

- a. $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$
- b. $\{w \mid w \text{ contains at least three 1s}\}$
- c. $\{w \mid w \text{ contains the substring 0101 (i.e., } w = x0101y \text{ for some } x \text{ and } y)\}$
- d. $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\}$
- e. $\{w \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$
- f. $\{w \mid w \text{ doesn't contain the substring 110}\}$
- g. $\{w \mid \text{the length of } w \text{ is at most 5}\}$
- h. $\{w \mid w \text{ is any string except 11 and 111}\}$
- i. $\{w \mid \text{every odd position of } w \text{ is a 1}\}$
- j. $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$
- k. $\{\epsilon, 0\}$
- l. $\{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$
- m. The empty set
- n. All strings except the empty string

1.18 Exercise A-n:

1.18a

$$\Sigma = \{0, 1\} \quad R = 1\Sigma^*0 = 1(0+1)^*0$$

10, 100, 110, 1010, 1100, 1010100

$$1(0+1)^*0$$

1.18b

R

$$\Sigma = \{0, 1\}$$

111, 010101, 011010000111, ...

$$R = \Sigma^*1\Sigma^*1\Sigma^*1\Sigma^*$$

$$= (0+1)^*1(0+1)^*1(0+1)^*1(0+1)^*$$

$$(0+1)^*1(0+1)^*1(0+1)^*1(0+1)^*$$

1.18c $\Sigma = \{0, 1\}$ - This is gonna be for every single one, unless otherwise specified

$$R = \Sigma^*0101\Sigma^* = (0+1)^*0101(0+1)^*$$

0101, 001011, 101011, 1101010, ...

$$(0+1)^*0101(0+1)^*$$

1.18d

$$R = \Sigma\Sigma^*0\Sigma^* = (0+1)(0+1)^*0(0+1)^*$$

000, 1101, ...

$$(0+1)(0+1)^*0(0+1)^*$$

1.18e

$$R = 0(\Sigma\Sigma^*)^* + 1(\Sigma\Sigma^*)^*$$

$$= 0((0+1)(0+1)^*)^* + 1((0+1)(0+1)^*)^*$$

0, 011, 010, 00101, 10, 11, 1001

$$0((0+1)(0+1)^*)^* + 1((0+1)(0+1)^*)^*$$

1.18f

$$R = 0^*(10^+)^*1^*$$

010, 011, 0101

$$0^*(10^+)^*1^*$$

1.18g

$$R = \Sigma + \Sigma + \Sigma\Sigma + \Sigma\Sigma\Sigma + \Sigma\Sigma\Sigma\Sigma + \Sigma\Sigma\Sigma\Sigma\Sigma$$

$$= \Sigma + (0+1) + (0+1)(0+1) + (0+1)(0+1)(0+1) + (0+1)(0+1)(0+1)(0+1) + (0+1)(0+1)(0+1)(0+1)(0+1)$$

+ $(0+1)^5$ + $\Sigma, 0, 01, 10, 1010, 00000, \dots$

$$= \Sigma + (0+1) + (0+1)^2 + (0+1)^3 + (0+1)^4 + (0+1)^5 + \frac{(0+1)^6}{(0+1)^4 + (0+1)^5}$$

$$1.18h \quad R = \Sigma + \Sigma + 0E + 10 + 0\Sigma + 10\Sigma + 110 + \Sigma^3 \Sigma^+ \\ \Sigma + (0+1) + (0+1) + 10 + 0(0+1)(0+1) + 10(0+1) + 110 + (0+1)^3(0+1) \\ \Sigma, 101, 110, 1010$$

$$\boxed{\Sigma + (0+1) + 0(0+1) + 10 + 0(0+1)(0+1) + 10(0+1) + 110 + (0+1)^3(0+1)}$$

$$1.18i \quad R = (\Sigma^+)^*(\Sigma+1) = (1(0+1))^*(\Sigma+1) \\ \Sigma, 101, 111, 1010$$

$$\boxed{(1(0+1))^*(\Sigma+1)}$$

$$1.18j \quad R = 00^*00^*(\Sigma+1) + 100^*(\Sigma+1)00^* + (\Sigma+1)00^*00^* \\ 001, 010, 100 \quad 00^* \\ 00^*00^*(\Sigma+1)$$

$$00^*00^*(\Sigma+1) + 00^*(\Sigma+1)00^* + (\Sigma+1)00^*$$

$$1.18k \quad L = \Sigma\Sigma, 0^2, \Sigma = \Sigma 0, 13 \\ R = 0 + \Sigma$$

$$\boxed{0 + \Sigma}$$

$$1.18l \quad R = 1^*(01^*01^*)^* + 0^*10^*10^*$$

$$\Sigma, 00, 11, 0101, 010100, \dots$$

$$\boxed{1^*(01^*01^*)^* + 0^*10^*10^*}$$

$$1.18m \quad L = \text{The empty set} \\ R = \emptyset$$

$$\boxed{\emptyset}$$

$$1.18 \quad R = \Sigma^* = (0+1)^*$$

$$\boxed{(0+1)^+}$$

RE to NFA Practice: 1.19a, 1.19b

1.19 Use the procedure described in Lemma 1.55 to convert the following regular expressions to nondeterministic finite automata.

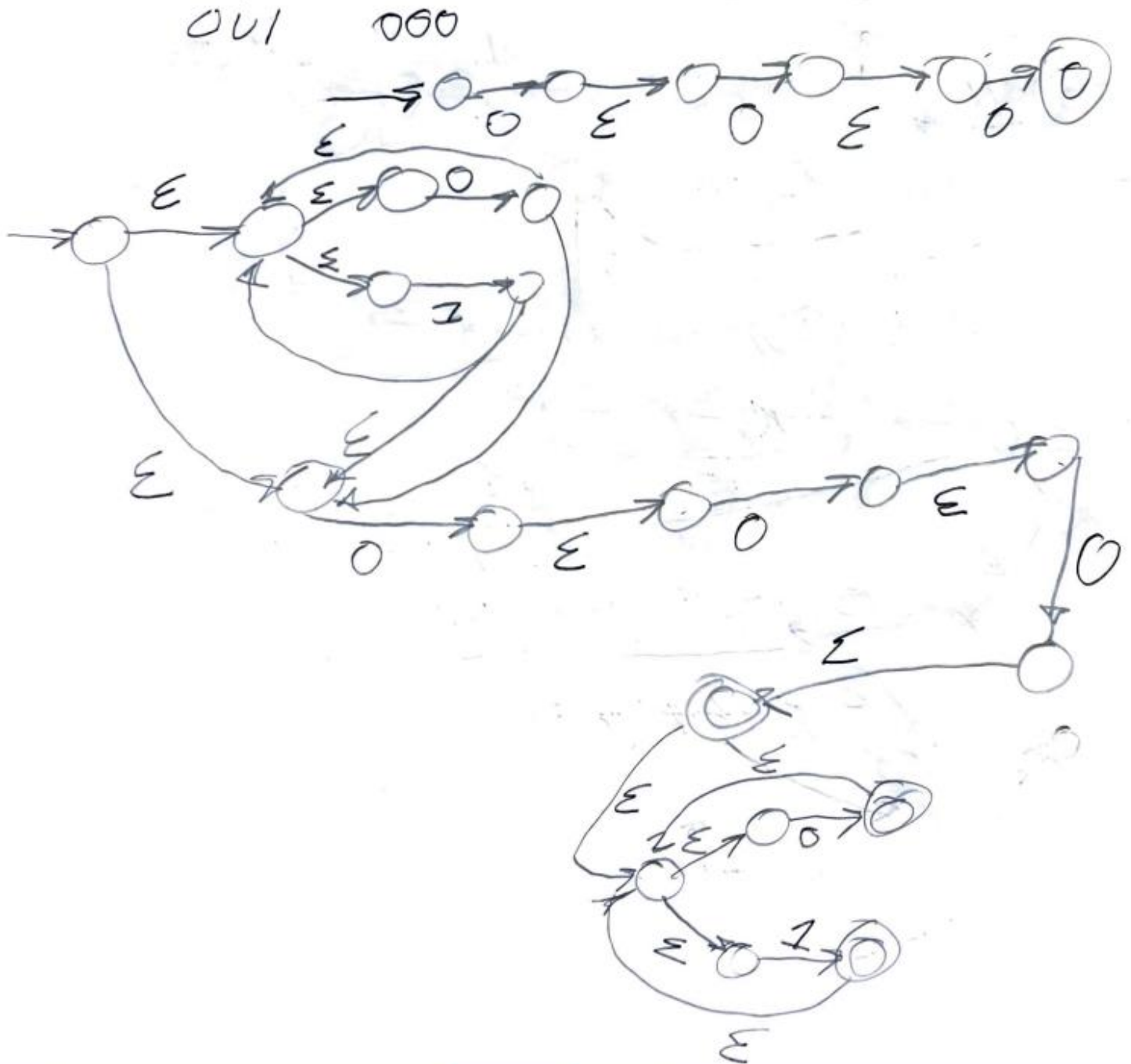
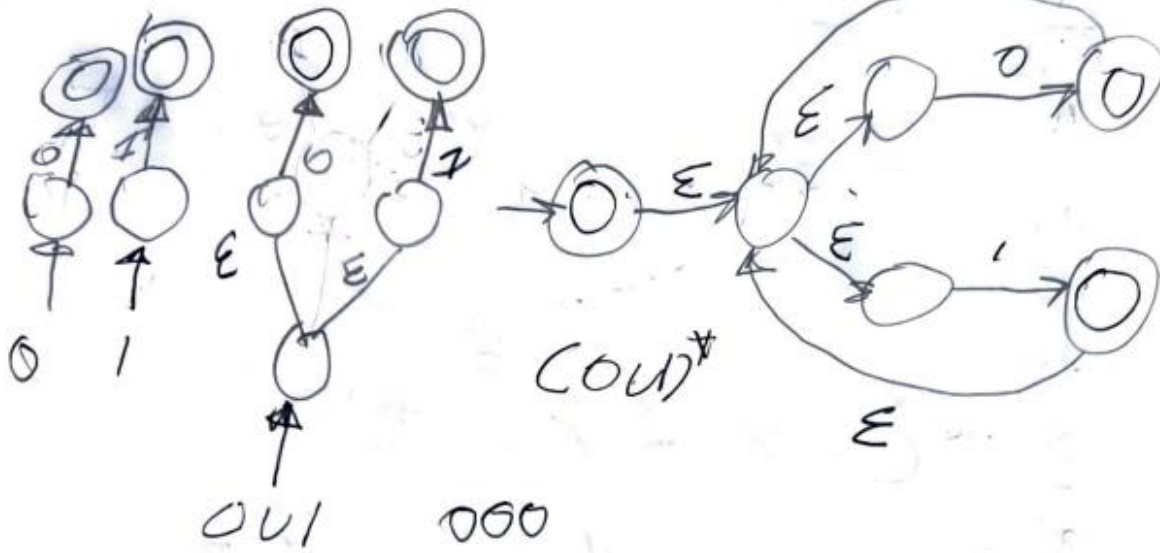
a. $(0 \cup 1)^* 000(0 \cup 1)^*$

b. $((00)^*(11) \cup 01)^*$

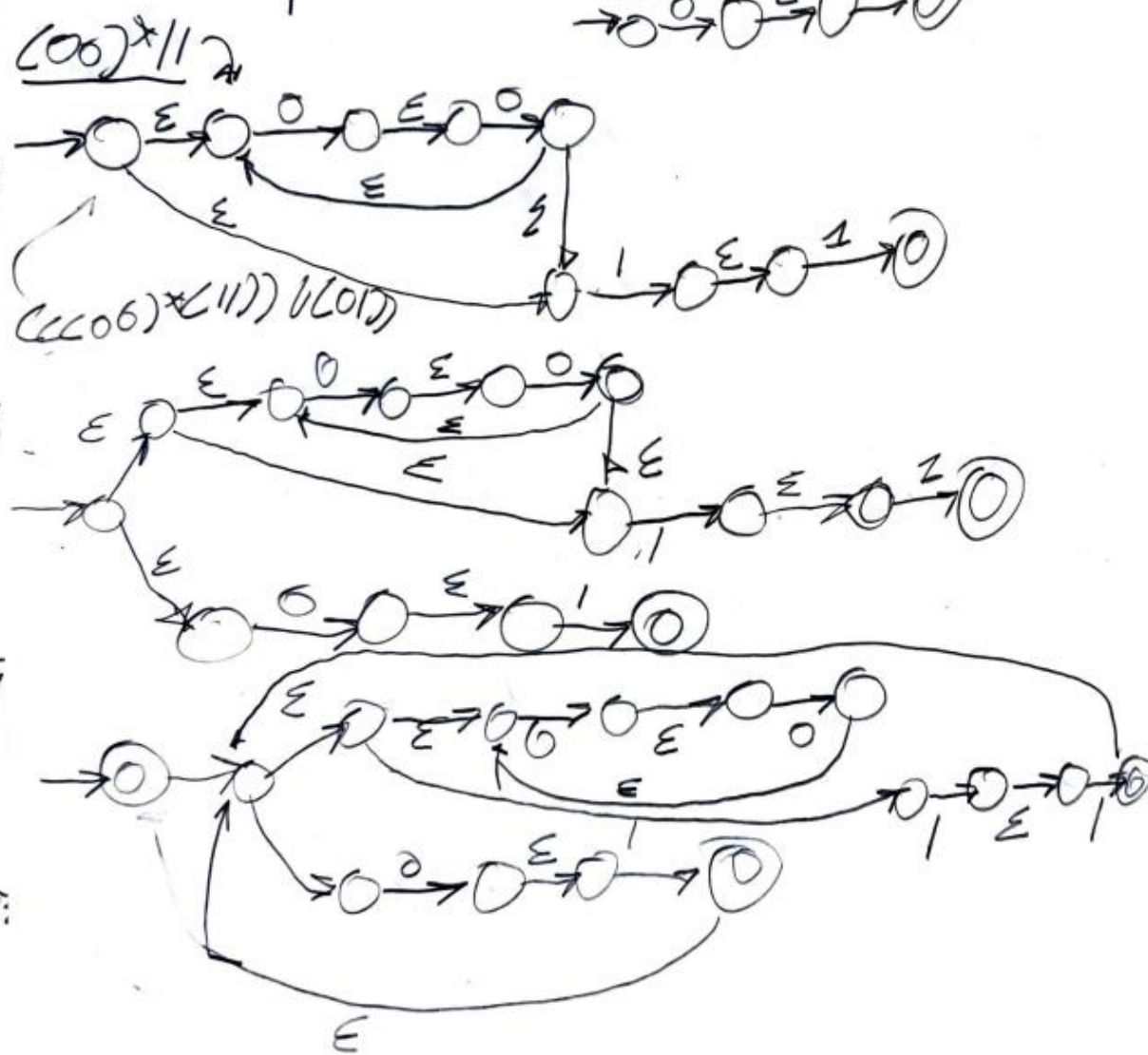
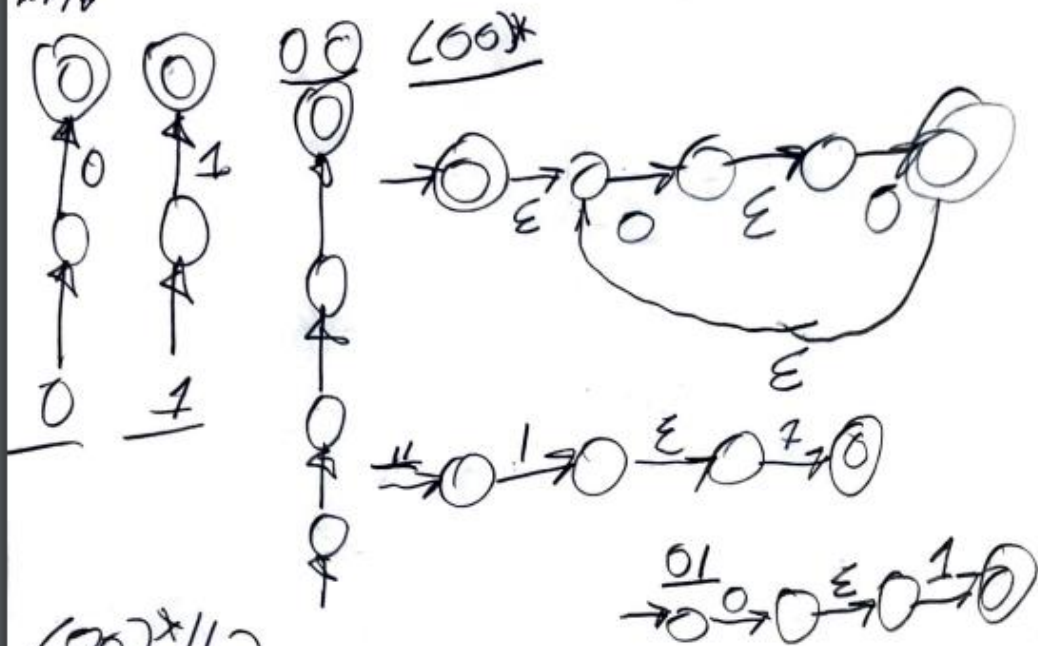
ended

1.190

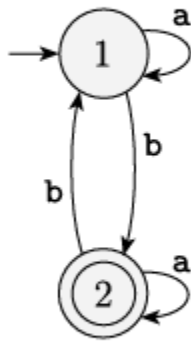
$$R = (0 \cup 1)^* 000 (0 \cup 1)^*$$



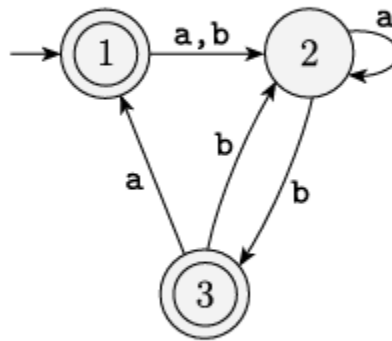
L196



1.21 Use the procedure described in Lemma 1.60 to convert the following finite automata to regular expressions.

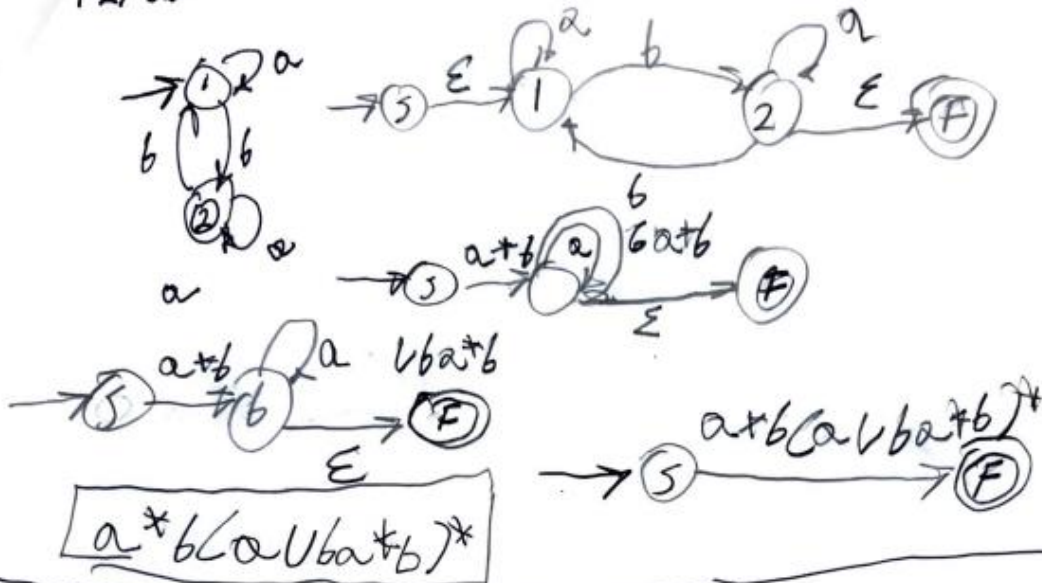


(a)

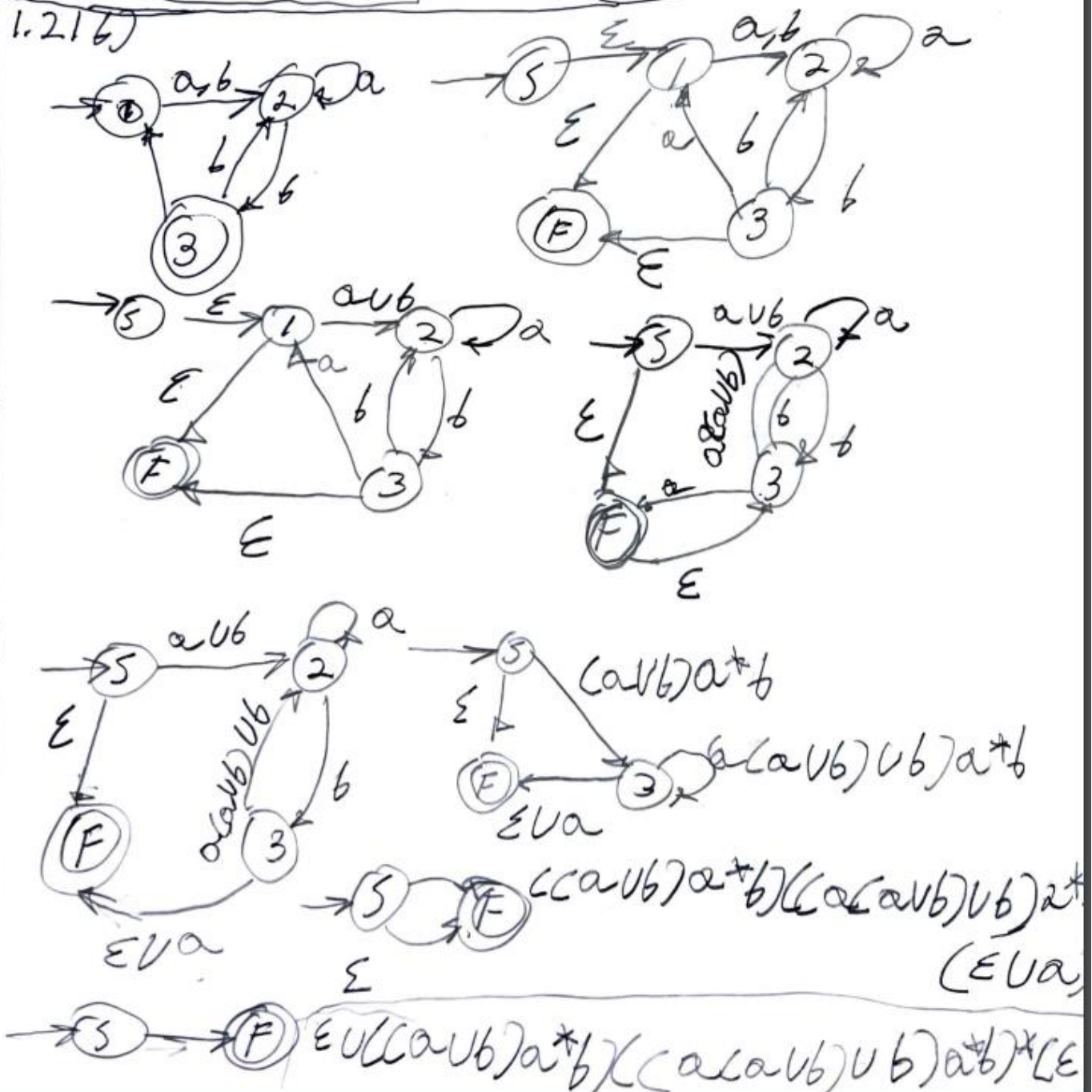


(b)

1.2) a)

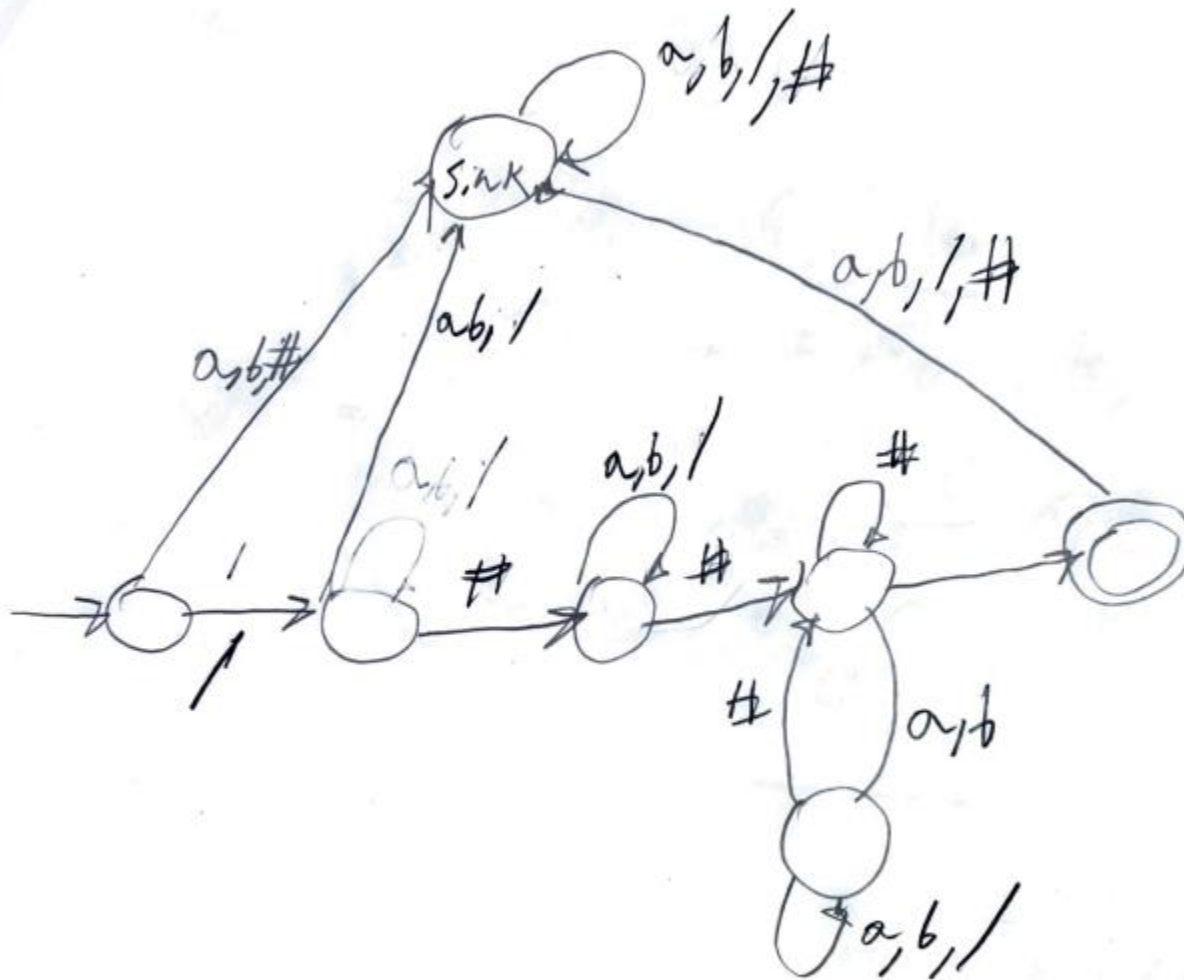


1.2) b)

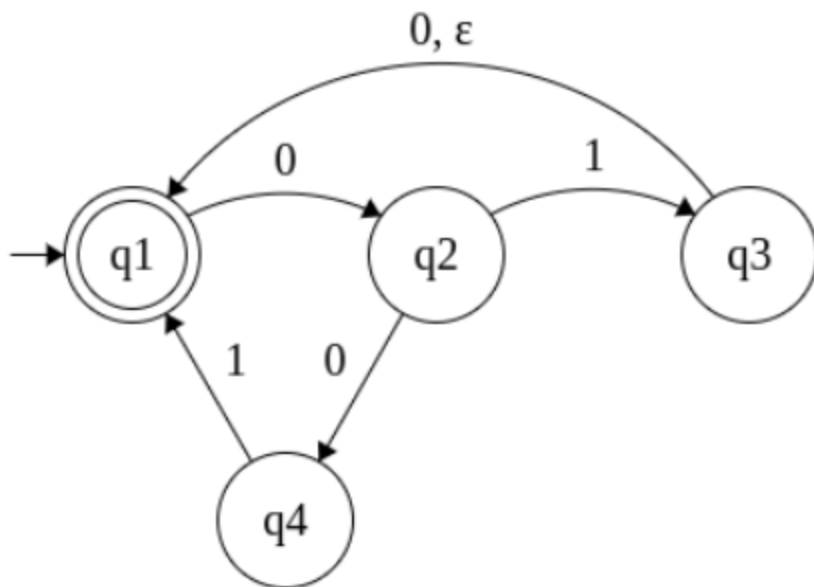


1.22 In certain programming languages, comments appear between delimiters such as $/\#$ and $\#/\$. Let C be the language of all valid delimited comment strings. A member of C must begin with $/\#$ and end with $\#/\$ but have no intervening $\#/\$. For simplicity, assume that the alphabet for C is $\Sigma = \{a, b, /, \#\}$.

- Give a DFA that recognizes C .
- Give a regular expression that generates C .



$/\#(a+b+/\#)^*\#(\#+(a+b)(a+b+/\#)^*\#)^*\#/\$

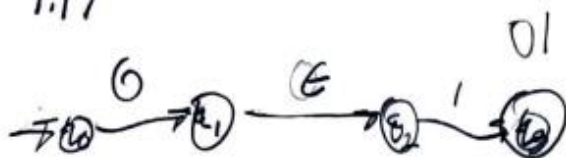


1.17

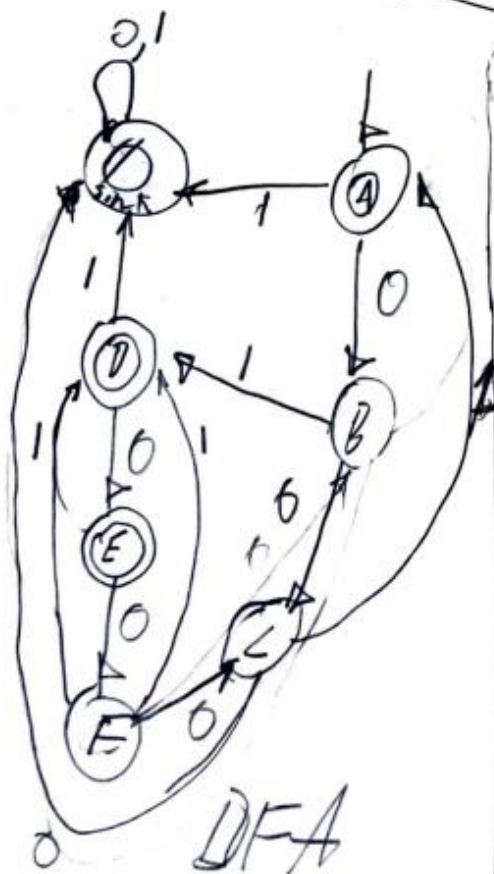
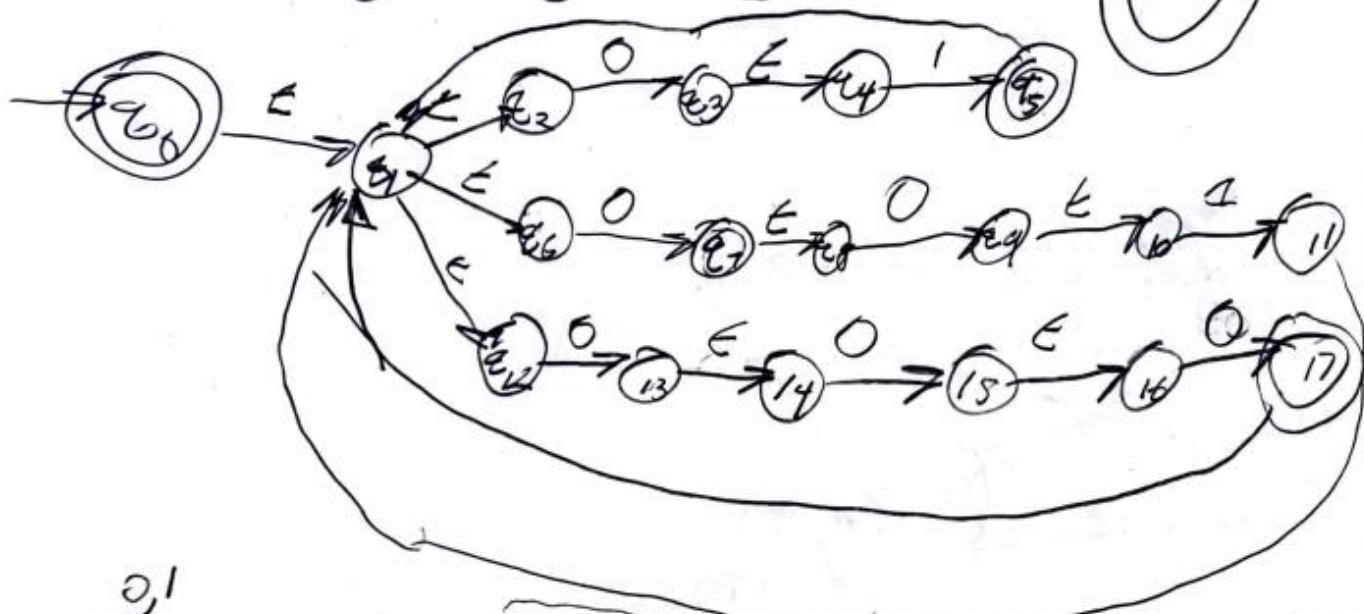
a. ~~Convert this NFA to an equivalent DFA. Give only the portion of the DFA that is reachable from the start state.~~

- b. Convert this NFA to an equivalent DFA. Give only the portion of the DFA that is reachable from the start state.

1.17



$L = (0^*10^*10^*)^*$



I'm a little confuse so 1.176 & next do NFA $\rightarrow$ DFA for the graph below? I think

1.176 wst The NFA Given in Class

